

# Metro Ticket Booking System

Technology Stack Specification (ServiceNow App Engine Architecture)

TARGET PLATFORM  
ServiceNow App Engine

DOCUMENT VERSION  
v1.0 (Tech Stack Specification)

APPLICATION SCOPE  
x\_snc\_metro\_book

## 1. Technical Stack Summary View

```
[ Front-End Layer ]    --->  Service Portal  |  Now Mobile App  |  Next Experience Workspace
                             |
[ Integration Layer ] --->  Integration Hub (Payments) |  Scripted REST APIs (Turnstiles)
                             |
[ Logic Engine ]       --->  Flow Designer   |  Script Includes   |  Scheduled Jobs
                             |
[ Data Storage ]       --->  sn_metro_ticket | sn_metro_station | sn_metro_payment
```

## 2. Presentation & User Experience Layer (Front-End)

| Component           | ServiceNow Technology                                                   | Purpose / Use Case                                                                                                     |
|---------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Commuter Web Portal | Service Portal & Service Catalog                                        | Responsive web portal for route searching, fare lookup, and ticket checkout.                                           |
| Custom UI Widgets   | Service Portal Widgets (HTML, CSS/ SASS, AngularJS, Client Controllers) | Interactive station selector, real-time ticket timer countdown, and embedded QR code canvas generator.                 |
| Mobile Experience   | Now Mobile / Mobile App Builder                                         | iOS/Android native mobile portal enabling commuters to purchase tickets on-the-go and display offline-cached QR codes. |
| Admin & Operations  | Next Experience Workspaces (UI Builder)                                 | Central workspace for station master management, fare matrix configuration, and transaction dispute handling.          |

## 3. Business Logic & Workflow Engine (Middle Tier)

| Component             | ServiceNow Technology                 | Purpose / Use Case                                                                                                               |
|-----------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Workflow Automation   | Flow Designer                         | Triggers post-payment workflows, coordinates multi-channel notifications (SMS/Email), and updates ticket status upon validation. |
| Server-Side Scripting | Script Includes (JavaScript / ES6)    | Reusable class library for fare calculation algorithms, station distance calculations, and cryptographic QR token signing.       |
| Event Drivers         | Business Rules (Async & Before/After) |                                                                                                                                  |

| Component         | ServiceNow Technology              | Purpose / Use Case                                                                                                      |
|-------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|                   |                                    | Validates ticket rules before record commits and triggers asynchronous status updates without blocking portal response. |
| Housekeeping Jobs | Scheduled Script Executions (Jobs) | Periodic background jobs that evaluate active tickets and transition past-expiry records from Active to Expired.        |

## 4. Integration & Hardware Gateway Layer

| Component        | ServiceNow Technology                           | Purpose / Use Case                                                                                                               |
|------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Gate Integration | Scripted REST API (Inbound Webhooks)            | Fast-responding inbound REST endpoint (/api/x_metro/gate_validation) processing turnstile scanner requests (<200ms latency).     |
| Payment Gateway  | Integration Hub (Outbound REST / Custom Spokes) | Outbound REST steps connecting to payment gateways (e.g., Stripe, Razorpay, UPI) to initiate transactions and receive callbacks. |
| Notifications    | Integration Hub / Twilio Spoke                  | Dispatches SMS receipts and WhatsApp/Push ticket notifications to commuters.                                                     |
| API Security     | Basic Auth / OAuth 2.0 / API Keys               | Secures turnstile hardware API calls and payment gateway webhook callbacks.                                                      |

## 5. Data Layer & Storage (Back-End Schema)

| Component        | ServiceNow Table               | Description & Schema Purpose                                                                                                                 |
|------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Station Master   | sn_metro_station               | Stores station names, station codes, line associations, and operational status.                                                              |
| Fare Matrix      | sn_metro_fare_matrix           | Distance-based or zone-based pricing lookup table between origin and destination pairs.                                                      |
| Ticket Records   | sn_metro_ticket (extends task) | Core ticket record holding unique ticket number, status (Draft, Active, In-Transit, Closed, Expired), valid window, and encrypted QR string. |
| Payment Ledger   | sn_metro_payment               | Audit table storing financial transaction IDs, payment methods, gateway responses, and refund references.                                    |
| Commuter Profile | sys_user                       | Core user table tracking commuter identity, contact info, and pass history.                                                                  |

## 6. Security, Access Control & Governance

- Role-Based Access Control (RBAC):
  - sn\_metro.user: Public commuter access to book tickets and view personal history.

- `sn_metro.gate_service`: Web service account restricted exclusively to turnstile API endpoints.
- `sn_metro.admin`: System administrator role for station matrices, fare configuration, and reports.
- **Data Encryption**: Column-Level Encryption / Encryption Contexts to protect sensitive payment tokens.
- **Scoped Application Isolation**: Encapsulated under scope `x_snc_metro_book` to prevent cross-scope platform interference.

## 7. Analytics & Operational Monitoring

- **Operational Reporting**: Real-time dashboards showing daily ticket revenue, peak travel hours, and top travel routes.
- **System Diagnostics**: Transaction logs tracking REST API response times and gate connectivity health.